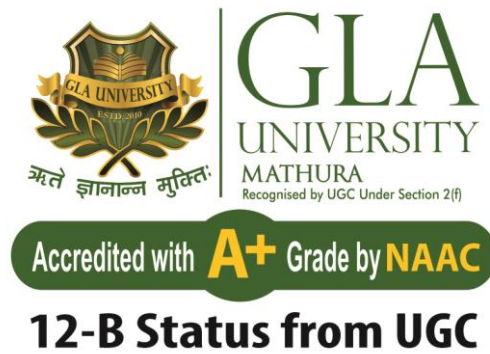


8-bit micro-processor

Architecture of 8085, Instructions, Instruction Sets & Addressing modes



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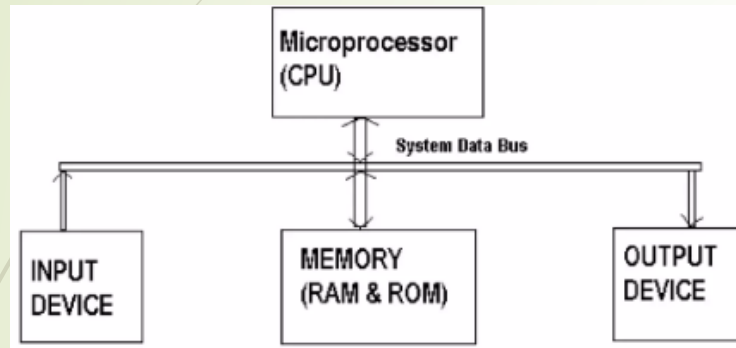


Outline

- Basics of Processor
- Features and Working
- Internal Architecture of 8085 Microprocessor
- Blocks/Groups of 8085: Registers, ALU, Timing & Control, Interrupt, serial I/O
- 8085 Languages
- 8085 Instructions: Format, classification & Instruction Set
- Addressing Modes

Basics of Processor

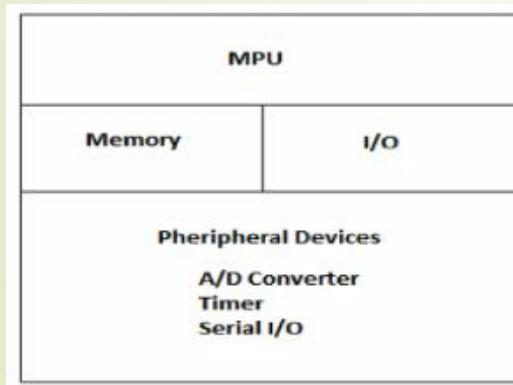
Microcomputer



Microcomputer:- It is a programmable machine. The two principal characteristics of a computer are:

- Responds to a specific set of instructions in a well-defined manner.
- It can execute a prerecorded list of instructions (a program)
- Its main components are
 - CPU
 - Input & Output devices
 - Memory

Microcontroller



- **Microcontroller:-** Silicon chip which includes microprocessor, memory & I/O in a single package

Microprocessor



Microprocessor:- It is a programmable VLSI chip which includes ALU, register circuits & control circuits. Its main units are-

- ALU
- Registers
- Control Unit

✓ Microprocessor is a **Programmable, Clock driven, Register based, Electronic device** that reads instruction from a storage device, takes the data from input unit and process the data according to the instructions and provides the result to the output unit.

- **Programmable-** Perform Different set operation on the data depending on the sequence of instructions supplied by the programmer.
- **Clock Driven** – Whole task is divided into basic operations, are divided into precise system clock periods.
- **Register Based** – Storage element
- **Electronic Device** – fabricated on a chip

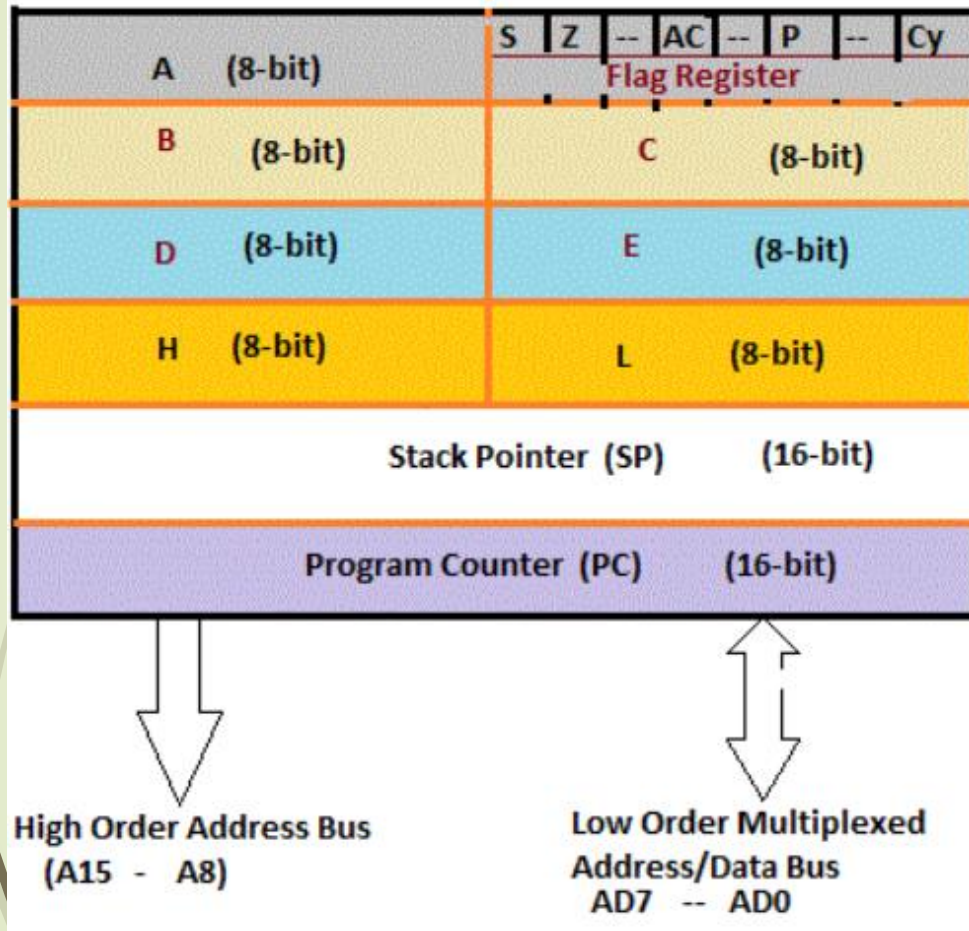
Features of 8085 μ p

- ✓ 8 bit microprocessor: can accept 8 bit data simultaneously
- ✓ Operates on single +5V D.C. supply.
- ✓ Designed using NMOS technology
- ✓ 6200 transistor on single chip
- ✓ It provides on chip clock generator, hence it does not require external clock generator.
- ✓ Operates on 3MHz clock frequency.
- ✓ 8bit multiplexed address/data bus, which reduce the number of pins.
- ✓ 16address lines, hence it can address $2^{16} = 64$ K bytes of memory
- ✓ It generates 8 bit I/O addresses, hence it can access $2^8 = 256$ I/O ports.
- ✓ 5 hardware interrupts i.e. TRAP, RST6.5, RST5.5, RST4.5, and INTR
- ✓ It provides DMA.

How does a microprocessor work?

- ✓ To execute a program, the microprocessor “reads” each instruction from memory, “interprets” it, then “executes or perform” it.
- ✓ The right name for the cycle is
 - Fetch
 - Decode
 - Execute
- ✓ This sequence is continued until all instructions are performed.

Programming Model of 8085 μ p

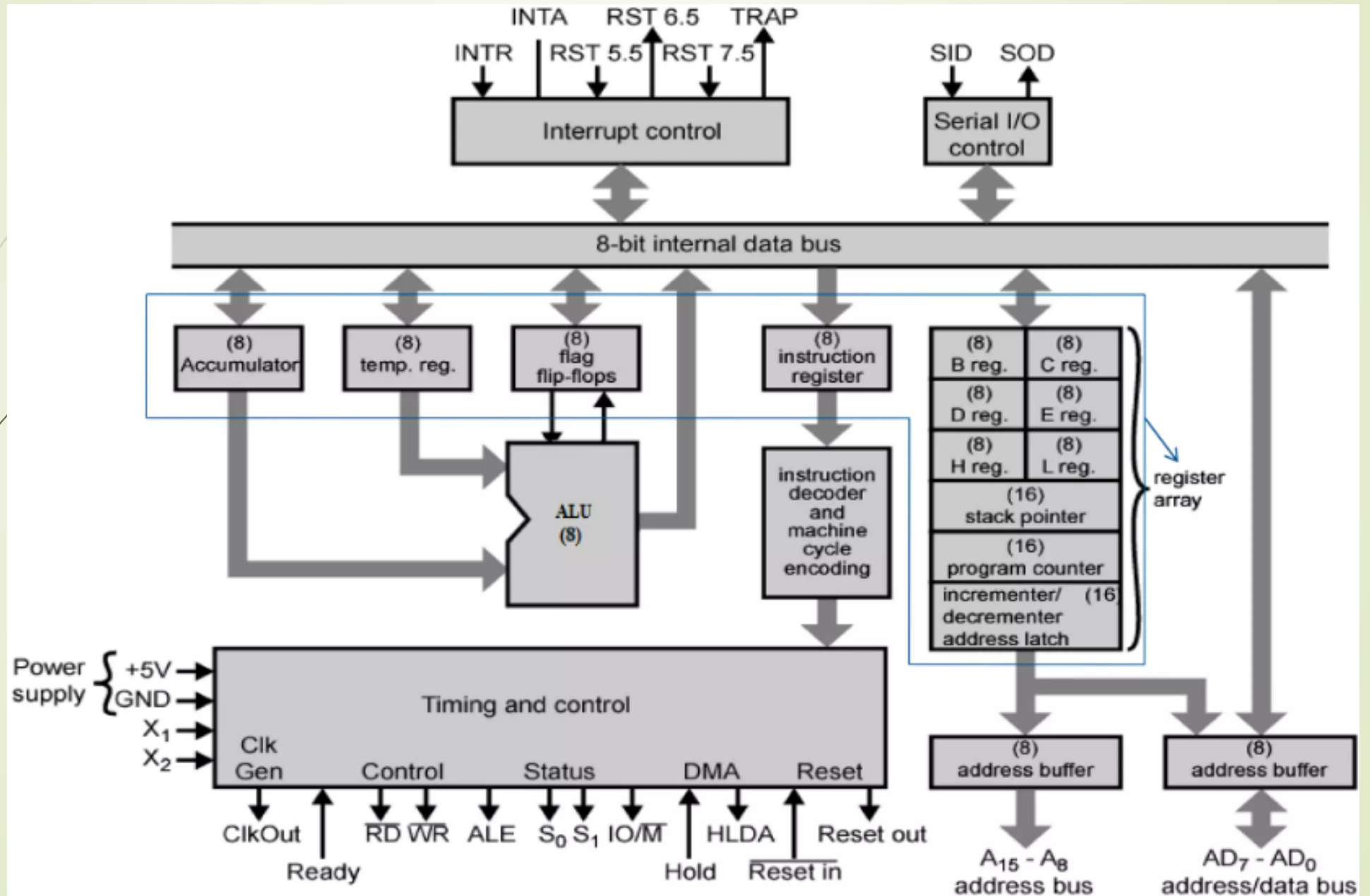


The programming model of 8085 consist of:

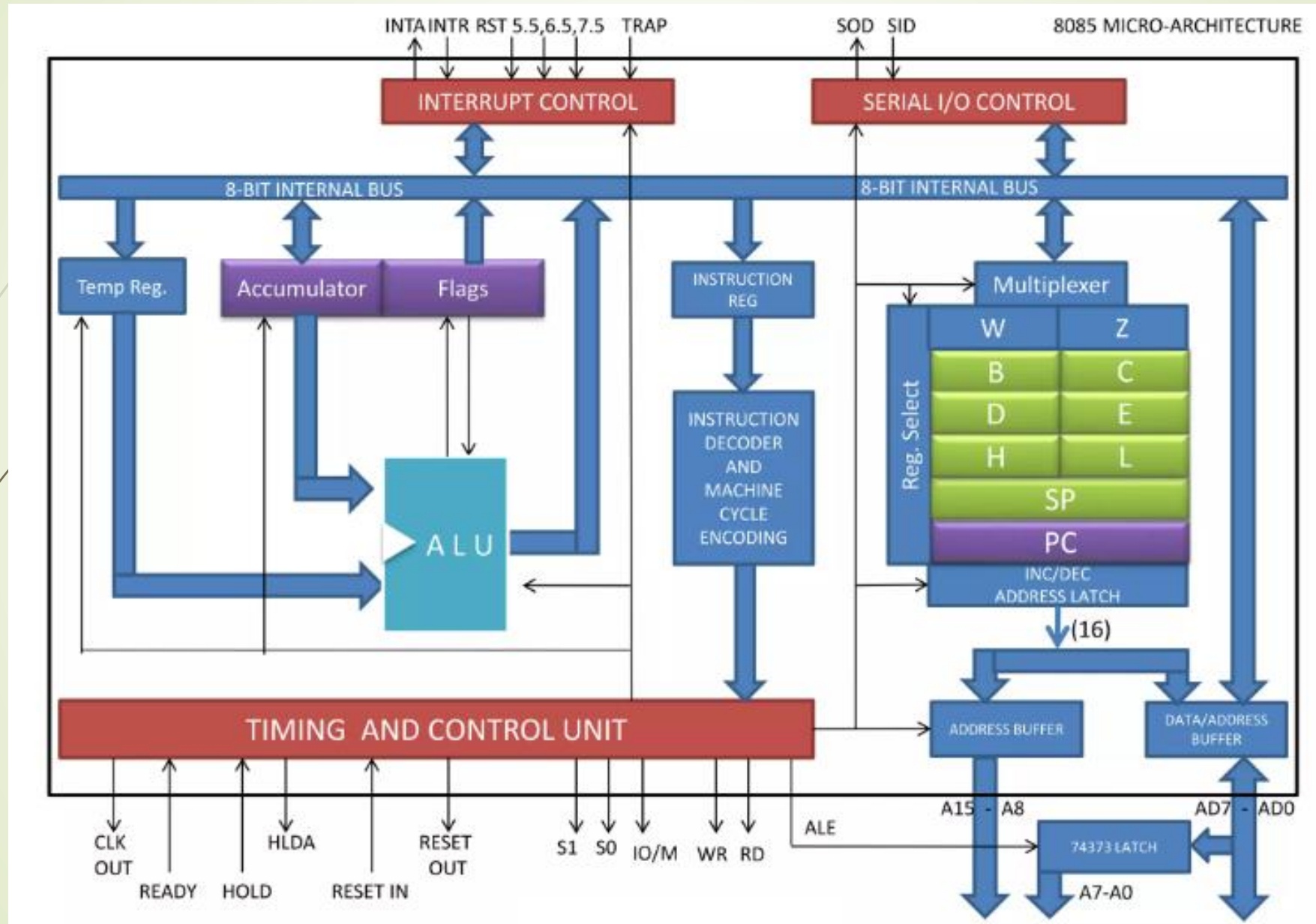
- An 8-bit accumulator
- One flag register
- Six general purpose register
- Two special purpose registers

****** These registers are critically required when programming a 8085 processor

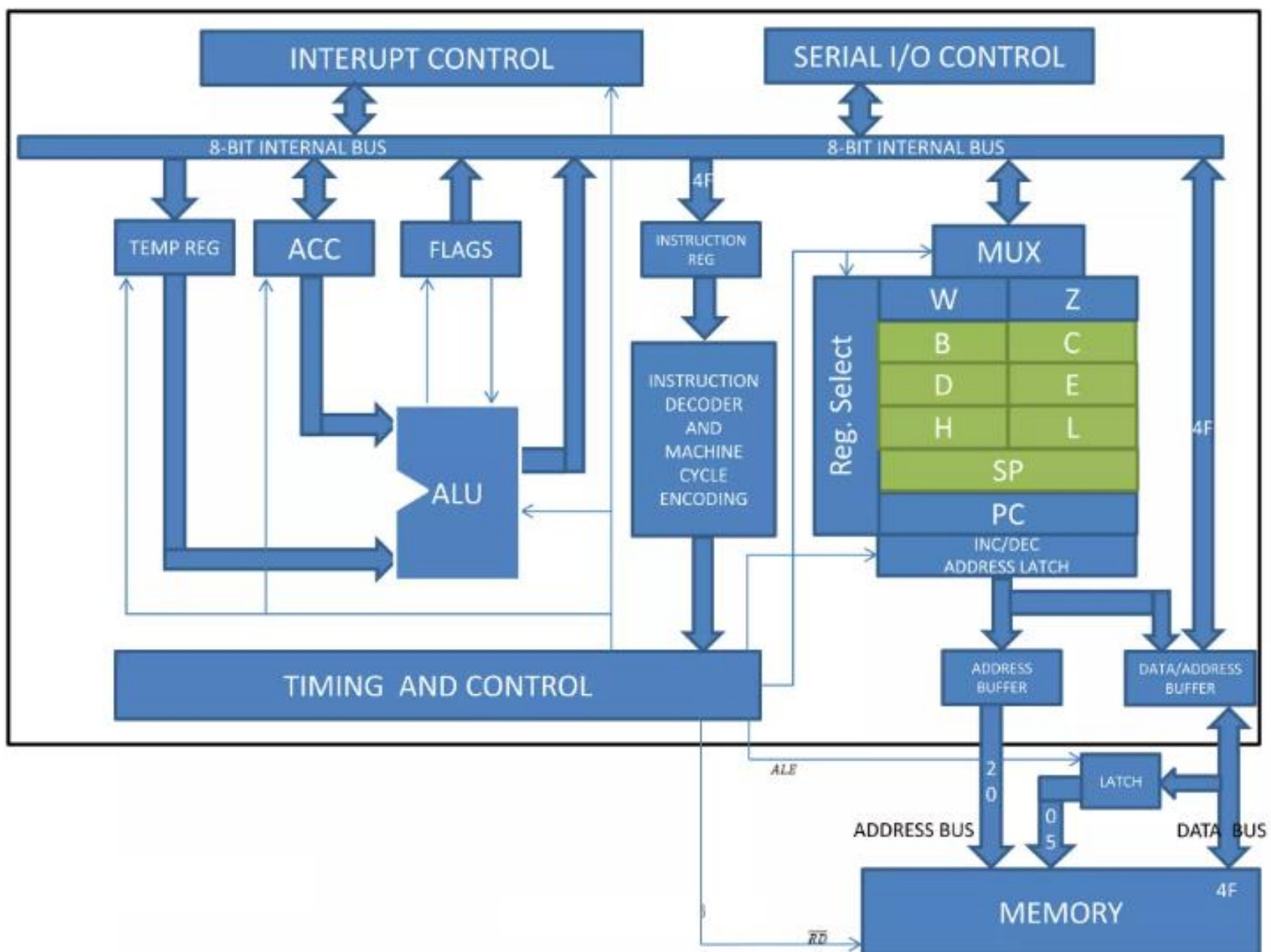
Internal Architecture of 8085



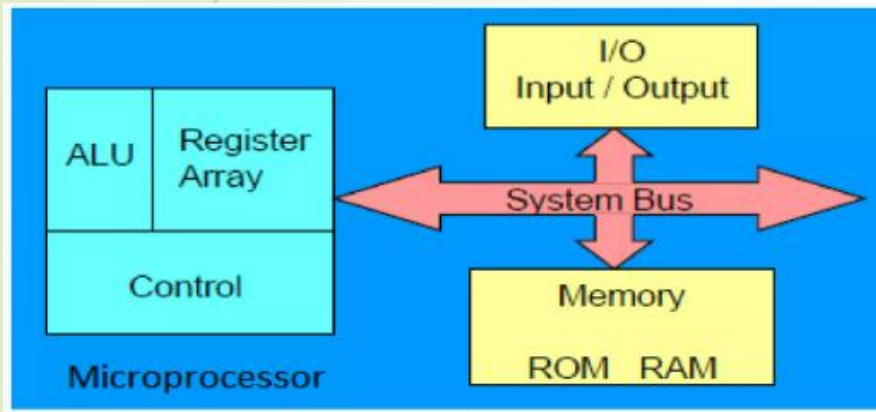
Internal Architecture of 8085



Internal Architecture of 8085



Microprocessor based system with BUS architecture



- **ALU:-** Arithmetic and logical operations like add, subtraction, AND & OR.
- **Register Array:-** Store data during the execution of program.
- **Control Unit:** Provides necessary timing & control signal. It controls the flow of data between microprocessor and peripherals.

• Memory:

- Stores information such as instructions and data in binary format (0 and 1).
- Sub-system” of microprocessor-based system. sub-system includes the registers inside the microprocessor .

✓ **Read Only Memory (ROM):** used to store programs that do not need alterations.

✓ **Random Access Memory (RAM) (R/WM):** used to store programs that can read and altered like programs and data.

- **Input/output:** Communicates with the outside world.
- **System Bus:** Communication path between the microprocessor and peripherals.
 - group of wires to carry bits.

Major Blocks of 8085 Microprocessor

The architecture consists of six blocks

1. Register arrays
2. ALU
3. Instruction decoder and machine cycle encoder, Timing and control circuit
4. Interrupt control
5. Serial I/O control

Registers of 8085

There are 14 registers, out of which 12 are of 8 bits and 2 are of 16 bit.
Registers are classified into four categories:

(a) **General purpose register: (user accessible)**

- B,C,D,E,H,L are 8 bit register.(can be used singly)
- Can also be used for 16-bit register pairs- BC, DE & HL.
- Used to store the intermediate data and result
- H & L can be used as a data pointer(holds memory address)

(b) **Special Purpose Register[A, Instruction Register and Flag]**

(b.1) **Accumulator (A): (user accessible)**

- 8 bit register
- All the ALU operations are performed with reference to the contents of Accumulator.
- Result of an operation is stored in A.
- Store 8 bit data during I/O transfer

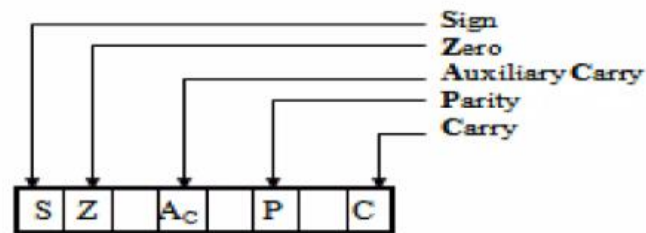
(b.2) **Instruction Register: (user not accessible)**

- When an instruction is fetched from memory, it is loaded in IR. Then transferred to the decoder for decoding.
- It is not programmable and can not be accessed through any instruction.

Registers of 8085 Cont...

(b.3) Flag Register(F): (user accessible)

- 8 bit Register
- Indicates the status of the ALU operation.
- ALU includes 5 flip flop, which are set or reset after an operation according to data conditions of the result in the accumulator.



(Flag Register)

Flag	Significance
C or CY (Carry)	CY is set when an arithmetic operation generates a carry out, otherwise it is 0 (reset)
P (Parity)	P= 1; if the result of an ALU operation has an even number of 1's in A; P= 0; if number of 1 is odd.
AC (Auxiliary carry)	Similar to CY, AC= 1 if there is a carry from D ₃ to D ₄ Bit AC= 0 if there is a no carry from D ₃ to D ₄ Bit (not available for user)
Z(zero)	Z = 1; if result in A is 00H 0 otherwise
S(Sign)	S=1 if D ₇ bit of the A is 1(indicate the result is -ive) S= 0 if D ₇ bit of the A is 0(indicate the result is +ive)

Registers of 8085 Cont...

(c) Temporary Register[W, Z, Temporary data register]

- Internally used by the MP(user not accessible)

(c.1) W and Z register:

- 8 bit capacity
- Used to hold temporary addresses during the execution of some instructions

(c.2) Temporary data register:

- 8 bit capacity
- Used to hold temporary data during ALU operations.

(d) Pointer Register or special purpose [SP, PC]

(d.1) Stack Pointer(SP)

- 16 bit register which holds the address of the data present at the top of the stack memory
- It is a reserved area of the memory in the RAM to store and retrieve the temporary information.
- Also hold the content of PC when subroutines are used.
- When there is a subroutine call or on an interrupt. ie. pushing the return address on a jump, and retrieving it after the operation is complete to come back to its original location.

(d.3) Program Counter(PC)

- 16 bit address used for the execution of program
- Contain the address of the next instruction to be executed after fetching the instruction it is automatically incremented by 1.
- Not much use in programming, but as an indicator to user only.

Registers of 8085 Cont...

In addition to register MP contains some latches and buffer

- ✓ **Increment and decrement address latch**
 - 16 bit register
 - Used to increment or decrement the content of PC and SP
- ✓ **Address buffer**
 - 8 bit unidirectional buffer
 - Used to drive high order address bus(A8 to A15)
 - When it is not used under such as reset, hold and halt etc this buffer is used tristate high order address bus.
- ✓ **Data/Address buffer**
 - 8 bit bi-Directional buffer
 - Used to drive the low order address (A0 to A7) and data (D0 to D7) bus.
 - Under certain conditions such as reset, hold and halt etc this buffer is used tristate low order address bus.

ALU of 8085

Arithmetic and logical group consists of (a) ALU (b) Accumulator (c) Temporary Registers (d) Flag Registers

(a) ALU

- Performs arithmetic and logical operations
- Stores result of arithmetic and logical operations in accumulator

(b) Accumulator

- General purpose register
- Stores one of the operand before any arithmetic and logical operations and result of operation is again stored back in Accumulator
- Store 8 bit data during I/O transfer

(c) Temporary Register

- 8 bit register
- During the arithmetic and logical operations one operand is available in A and other operand is always transferred to temporary register

For Eg.: ADD B – content of B is transferred into temporary register before actual addition

(d) Flag Register

- Five flag is connected to ALU
- After the ALU operation is performed the status of result will be stored in five flags.

Architecture of 8085 : Group 3, 4 & 5

Instruction decoder and machine cycle encoder

- Decodes the op-code stored in the Instruction Register (IR) and establishes the sequence of events to follow.
- Encodes it and transfer to the timing & control unit to perform the execution of the instruction.

Interrupt control

- **Interrupt:-** Occurrence of an external disturbance
- After servicing the interrupt, 8085 resumes its normal working sequence
- Transfer the control to special routines
- Five interrupts: - TRAP, RST7.5, RST6.5, RST5.5, INTR
- In response to INTR, it generates INTA signal

Timing and control circuit

- works as the brain of the CPU
- For proper sequence and synchronization of all the operations of MP, this unit generates all the timing and control signals necessary for communication between microprocessor and peripherals.

Serial I/O control

- Data transfer red on Do- D7 lines is parallel data
- But under some condition it is used serial data transfer
- Serial data is entered through SID(serial input data) input (received)
- Serial data is outputted on SOD(serial output data) input (send)

8085 Language Languages

Machine Language

- ✓ To communicate with computer, instruction is given in binary language.
- ✓ MP has 8 bit data so $2^8 = 256$ combinations. So difficult to write programs in set of 0's and 1's.

for eg:

- 0011 1100 – is an instruction that increments the number in accumulator by 1
- 1000 0000– is an instruction that add the number in register B to the accumulator content , and keep the result in A.

So it is very tedious and error inductive. For convenience, written in Hexadecimal code. For example 0011 1100 is equivalent to 3C H

Assembly Language

- ✓ Even program is written in Hexa decimal .. It is difficult to understand.
- ✓ Program is written in mnemonic.

For E.g.: binary code 0011 1100 (3C H in hexadecimal) is represented by INR A

- INR A –INR stands for Increment, A is accumulator... this symbol suggest the operation of incrementing the accumulator by 1

Similarly 1000 0000 is equivalent (80 H) is represented as

- ADD B– ADD stands for addition and B represents content in register B. this symbol suggest that addition of the number in register B to the accumulator content , and keep the result in A.

***So MP has 246 such bit pattern amounting to 74 different instruction for performing various operations

8085 Instructions

❑ Since the 8085 is an 8-bit device it can have up to 28 (256) instructions.

- However, the 8085 only uses 246 combinations that represent a total of 74 instructions.
- Most of the instructions have more than one format.

❑ These instructions can be grouped into five different groups

- Data Transfer Operations
- Arithmetic Operations
- Logic Operations
- Branch Operations
- Machine Control Operations

Instructions Formats

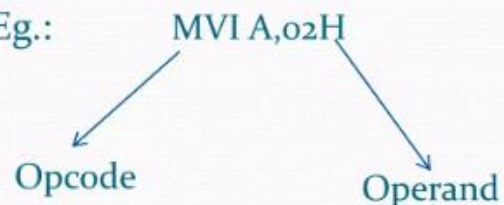
❑ Each instruction has two parts.

❑ **Opcode (operation code)**- The first part is the task or operation to be performed.

❑ **Operand** - The second part is the data to be operated on. Data can be given in various form.

- It can specify in various ways: may include 8 bit/16 bit data, an internal register, memory location or 8 bit /16 bit address.

❑ For Eg.:



Instruction Word Size

➤ An instruction is assembled in binary form (0,1) known as machine code or opcode. Due to different ways of specifying data or operand the machine code are not same for all the instruction

➤ The size of an instruction signifies how much memory space is required to load an instruction in the memory. 8085 instructions are of following sizes:

✓ **One-byte or one word Instructions:** opcode and operand in 8 bits only i.e. one byte. Operand(s) are internal register and are coded into the instruction.

e.g. MOV, ADD, ANA, SUB, ORA etc.

MOV A,B Move the content of B in A
01 111 000 = 78H 01 for MOV operation, 111 binary code for register A and last 000 binary code for register B

Task	Op- code	Operand	Binary Code	Hex Code
Copy the contents of the accumulator in the register C.	MOV	C,A	0100 1111	4FH
Add the contents of register B to the contents of the accumulator.	ADD	B	1000 0000	80H
Invert (compliment) each bit in the accumulator.	CMA		0010 1111	2FH

✓ **Two-byte instructions:** first byte is opcode in 8 bits and second byte is operand either 8 bit data or 8 bit address.

e.g. MVI, ADI, ANI, ORI, XRI etc.

MVIB, 05H Move 05 in register B
06,05 MVIB, 05H in the code for m

Task	Op- code	Operand	Binary Code	Hex Code
Load an 8-bit data byte in the accumulator.	MVI	A, Data	0011 1110	3E (First Byte)
			DATA	Data(Second Byte)

Mnemonics	Hex code
MVI A, 32H	3E 32H

Instruction Word Size Cont...

✓ **Three-byte instructions:** first byte is opcode in 8 bits and second and third byte are operand either 16 bit data or 16 bit address.

Operand 1 = lower 8 bit data/address

Operand 2 = Higher 8 bit data/address

opcode + data byte + data byte

e.g. LXI, LDA, STA, LHLD, SHLD etc.

LXIH , 2500H
21,00,25

load HL pair with 2500H
LXIH , 2500H in code form

Task	Op-code	Operand	Binary Code	Hex Code	Hex Code
Transfer the program sequence to the memory Location 2085H.	JMP	2085H	1100 0011	C3	First Byte
			1000 0101	85	Second Byte
			0010 0000	20	Third Byte

8085 Instructions Set

- ✓ Since the 8085 is an 8-bit device it can have up to 28 (256) instructions.
- ✓ However, the 8085 only uses 246 combinations that represent a total of 74 instructions.
- ✓ Most of the instructions have more than one format.

These instructions can be grouped into five different groups:

- Data Transfer Operations
- Arithmetic Operations
- Logic Operations
- Branch Operations
- Machine Control Operations

Data Transfer operation

- These operations simply **COPY** the data from the source to the destination.
- They transfer:
 - Data between registers.
 - Data Byte to a register or memory location.
 - Data between a memory location and a register.
 - Data between an I/O Device and the accumulator.
- The data in the source is not changed.
- Data transfer instructions never affect the flag bits.
e.g. LDA, STA, MOV, LDAX, STAX, MVI, LXI etc.

8085 Instructions Set Cont...

Arithmetic Operation

- These instructions perform addition, subtraction and compare operations.
- These operations are always performed with accumulator as one of the operands.
- The status of the result can be verified by the contents of the flag register.

Addition: Any 8-bit number, or the contents of a register or the contents of a memory location can be added to the contents of the accumulator and the sum is stored in the accumulator. The instruction **DAD** is an exception; it adds 16-bit data directly in register pairs. Ex ADD, ADI

Subtraction - Any 8-bit number, or the contents of a register, or the contents of a memory location can be subtracted from the contents of the accumulator and the results stored in the accumulator. Subtraction is done by 2's complement method and set carry flag to indicate borrow. SUB, SBI

Increment/Decrement - The 8-bit contents of a register or a memory location can be incremented or decremented by 1. Similarly, the 16-bit contents of a register pair (such as BC) can be incremented or decremented by 1. INR, DCR.

Logical Operation

- Perform 8-bit basic logical operations with the content of the accumulator
- Logical instructions also modify the flag bits.
- Op-codes for logical instructions include ANA, ANI, ORA, ORI, XRA, XRI, CMA, CMC, RAL, RLC, RAR, RRC, CMP, CPI etc.

AND, OR Exclusive-OR - Any 8-bit number, or the contents of a register, or of a memory location can be logically AND, Or, or Exclusive-OR with the contents of the accumulator. The results are stored in the accumulator.

Rotate- Each bit in the accumulator can be shifted either left or right to the next position.

Compare- Any 8-bit number, or the contents of a register, or a memory location can be compared for equality, greater than, or less than, with the contents of the accumulator.

Complement - The contents of the accumulator can be complemented.

8085 Instructions Set Cont...

Branch Operation

These instructions are used to transfer the program control:

- To jump from one memory location to any other memory location within a program
- From one program to another program called as a subroutine.
- Alters the sequence of program execution either conditionally or unconditionally
- **Unconditional branch instructions**- Transfer the program to the specified label or address JMP unconditionally i.e. without satisfying any condition.

Unconditional Program control instructions are

- Call & RET

- **Conditional branch instructions** -Transfer the program to the specified label or address when certain condition is satisfied.
 - JNC, JC, JNZ, JZ, JP, JM, JPE, JPO
 - CNC, CC, CNZ, CZ, CP, CM, CPE, CPO
 - RNC, RC, RNZ, RZ, RP, RM, RPE, RPO

Machine Control Operation

These instructions include special instructions such as I/O data transfer, perform machine related operation

- HLT - To halt the CPU
- NOP - To perform no operation
- SIM - To set the masking of hardware interrupts and serial output data
- RIM - To read the status of interrupt mask and serial input data
- EI - Enable Interrupt
- DI - Disable Interrupt

Addressing Modes

The various formats for specifying operands/data are called the ADDRESSING MODES. 8085 instructions can be classified in following addressing modes

❑ Register Addressing mode

- Data is provided through the registers .e.g. MOV, ADD, SUB, ANA, ORA, XRA etc.
For Ex. MOV A, B move the content of register B to register A.
 78H opcode of MOV A,B

❑ Immediate Addressing mode

- Data is present in the instruction. Load the immediate data to the destination provided. e.g. MVI, LXI, ADI, SUI, ANI, ORI etc.
For Ex. MVI A, 05H move the 05H in register A.
 3E, 05 opcode of MVI A, 05H

❑ Direct Addressing mode

- Instructions have their operands in memory and the 16-bit memory address is specified in the instruction
- The address of the operand is given in the instruction. e.g. LDA, STA, LHLD, SHLD etc.
For Ex.
 STA 7500 store the content of accumulator in the memory location 7500 H
 32 , 00, 75 opcode of STA 7500

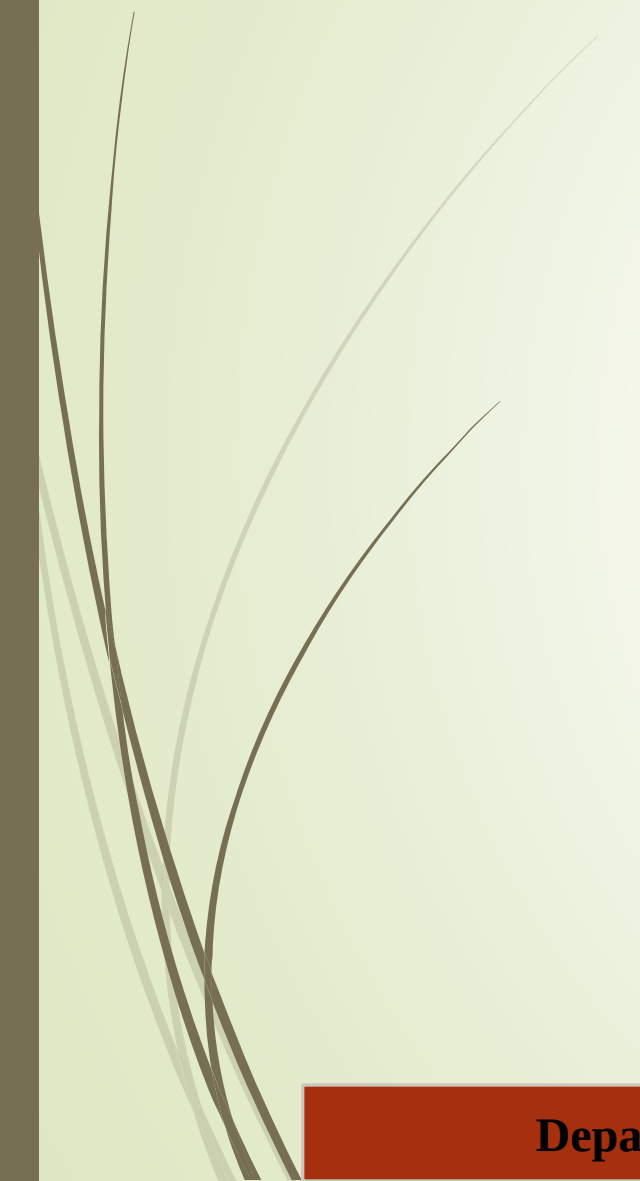
 IN 01 Read data from port 01.
 DB,01 opcode of IN 01

❑ Indirect Addressing mode

- In this address of the operand is specified by the register pair. The address stored in the register pair points to memory location e.g. LDAX, STAX, PUSH, POP etc.
 LXI H, 7500 load HL pair with 7500H
 MOV A, M move the content of memory location,
 whose address is in HL pair to the accumulator

❑ Implicit or Implied Addressing mode

- The operand is not specified in the instruction, specified within the opcode itself.
- Operand is supposed to be present in accumulator. e.g. CMA, CMC, STC etc.
 RAL Rotate the content of accumulator towards left.



Thanks